银发智慧与科学生辉长者生命科学互动沙龙活动

提案

一、 实施背景与总体目标

随着前沿生命科学与医疗诊断技术的快速发展，老年群体在获取并理解新型公共卫生技术信息时面临客观的认知落差。本项目的核心目标为消除此类信息壁垒，将合成生物学的底层逻辑与呼吸道疾病快速检测技术转化为老年群体可直接感知的实体形式。同时，建立项目研发团队与老年用户的双向沟通渠道，系统收集该群体对便携式医疗设备在适老化设计与健康管理方面的实际需求。

二、 目标受众与场地标准

受众群体：康养机构常住老年人及老年大学在籍学员。场地标准：须选取具备完善无障碍设施的社区活动中心或养老机构多功能厅。室内须满足自然采光充足、无强光眩目的视觉要求，配备的座椅必须具备防滑基座、双侧扶手及符合人体工学的适老背靠结构。

三、 核心执行程序

第一阶段：传统生物工艺与现代生物技术原理对比解析以日常酸奶发酵及传统酿酒工艺为切入点，客观陈述微生物代谢运作的基础现象。随后依据此逻辑，递进讲解现代合成生物学如何通过基因序列重组实现对微生物代谢路径的精确干预与目标产物控制。

第二阶段：宏观适老化生物学物理模型触觉感知向参与者提供按特定比例放大的脱氧核糖核酸双螺旋结构实体模型及细胞内部结构物理模型。引导伴有视力衰退特征的长者通过手部触摸与形态感知，建立对微观生命科学底层结构的具象认知。

第三阶段：呼吸道健康与生物传感技术专题宣讲结合老年群体免疫机能特征，开展呼吸道传染病预防的专业健康宣讲。同步向长者展示便携式生物传感器的结构组成、病原体识别原理及其在基层医疗干预与日常健康监测中的应用路径。

第四阶段：医疗技术演进历程双向经验采录邀请参与活动的适龄长者基于自身生命历程，口述其所经历的医疗诊断技术与公共卫生服务体系演进过程。项目团队须同步记录长者对新型分子检测设备的接受意愿、操作顾虑及产品迭代建议。

四、 适老化保障与风险控制机制

感官代偿机制：所有下发的宣教手册及展示文稿均须采用高对比度色彩搭配及特大号非衬线字体排版。宣讲现场须配备全频段扩音设备，并依据参与者的听力衰退程度动态调节输出分贝。医疗与人员应急保障：活动开展全过程须强制要求属地机构的全科医师或专业护理人员在场待命。项目团队须执行分区定点看护制度，保障每位参与者在活动落座、离席及接触物理模型环节的绝对人身安全。

Activity Proposal for Senior Life Science

Interactive Salon: Silver Wisdom and Scientific

Radiance

I. Implementation Background and Overall Objectives

With the rapid development of cutting-edge life sciences and medical diagnostic technologies, the elderly population faces an objective cognitive gap in accessing and understanding new public health technology information. The core objective of this project is to eliminate such information barriers and translate the underlying logic of synthetic biology and rapid respiratory disease detection technologies into physical forms that are directly perceivable by the elderly demographic. Simultaneously, it aims to establish a two-way communication channel between the project research and development team and elderly users, systematically collecting the actual demands of this group regarding the senior-friendly design of portable medical devices and health management.

II. Target Audience and Venue Standards

Target Audience: Long-term elderly residents in nursing care institutions and registered students at senior universities.Venue Standards: Community activity centers or multi-function halls in elderly care institutions equipped with comprehensive barrier-free facilities must be selected. The interior must meet the visual requirements of sufficient natural lighting without glaring reflections, and the provided seating must feature non-slip bases, bilateral armrests, and ergonomic senior-friendly backrest structures.

III. Core Execution Procedures

Phase One: Comparative Analysis of Traditional Biological Processes and Modern Biotechnology Principles

Starting with daily yogurt fermentation and traditional brewing processes, the basic phenomena of microbial metabolic operations will be objectively stated. Following this logic, a

progressive explanation will be provided on how modern synthetic biology achieves precise intervention in microbial metabolic pathways and target product control through genetic sequence recombination.

Phase Two: Tactile Perception of Macroscopic Senior-Friendly Biological Physical Models

Physical models of the deoxyribonucleic acid double helix structure and internal cell structures, scaled up to specific proportions, will be provided to the participants. Elders with characteristics of visual decline will be guided to establish concrete cognition of microscopic life science underlying structures through manual touching and morphological perception.

Phase Three: Special Presentation on Respiratory Health and Biosensing Technologies

Integrating the immune function characteristics of the elderly population, professional health presentations on the prevention of respiratory infectious diseases will be conducted. Concurrently, the structural composition of portable biosensors, the principles of pathogen identification, and their application pathways in grassroots medical interventions and daily health monitoring will be demonstrated to the elders.

Phase Four: Two-Way Experience Recording of the Evolution of Medical Technologies

Eligible elders participating in the activity will be invited to orally recount the evolution of medical diagnostic technologies and public health service systems they have experienced throughout their lives. The project team must synchronously record the elders' willingness to accept new molecular testing devices, their operational concerns, and their suggestions for product iteration.

IV. Senior-Friendly Guarantees and Risk Control Mechanisms

Sensory Compensation Mechanisms: All distributed educational manuals and presentation materials must utilize high-contrast color matching and extra-large sans-serif typography. The presentation venue must be equipped with full-band public address equipment, dynamically adjusting the output decibels according to the participants' degree of hearing decline.Medical and Personnel Emergency Guarantees: Throughout the entire duration of the activity, general practitioners or professional nursing staff from local institutions are mandatorily required to be on standby at the venue. The project team must execute a zoned and fixed-point caretaking system to ensure the absolute personal safety of each participant during seating, departure, and contact with physical models.